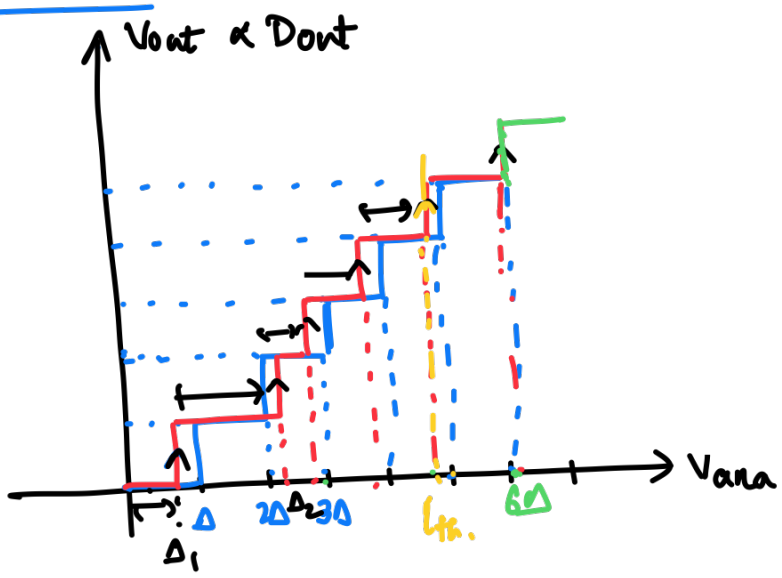
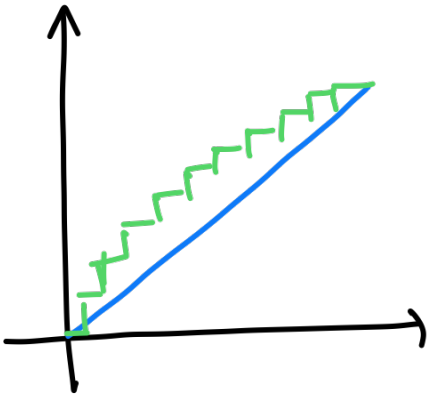
Lecture #39



- Gain error
- offset error



first step size = $\frac{\Delta_1 - 0}{\Delta}$ LSB
(Normalized)

2nd. = $\frac{\Delta_2 - \Delta_1}{\Delta}$

$DNL(i) = \frac{\text{Step size } (i) - \Delta}{\Delta} 1 =$ Normalize step size $- 1$

$-0.5 < DNL(i) < 0.5$

$INL(i) = \sum_{i<1}^{K} DNL(i)$

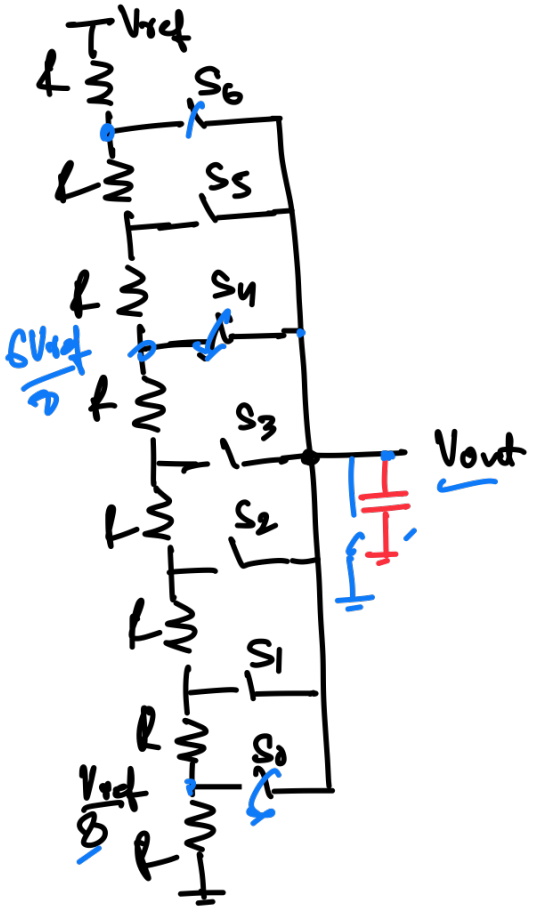
$= \sum_{i=1}^{IC}$ Normalized step size$_i$ $- IC.$

DAC : Digital-to-Analog Converter



DAC

- 3-bit DAC → Vout



- $3'b000 - 3'b111 \rightarrow \{S_0, S_1 \ldots S_6\}$

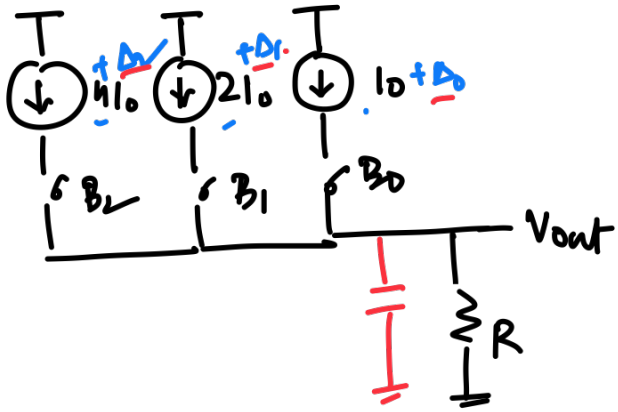
$3'b000 - 0$

$3'b111 - \frac{7V_{ref}}{8}$

-

- Current DAC (CDAC)

- Binary -



$V_{out} = I_0 R \left(B_0 + 2B_1 + 4B_2 \right) + 8B_3$

ΔI_0

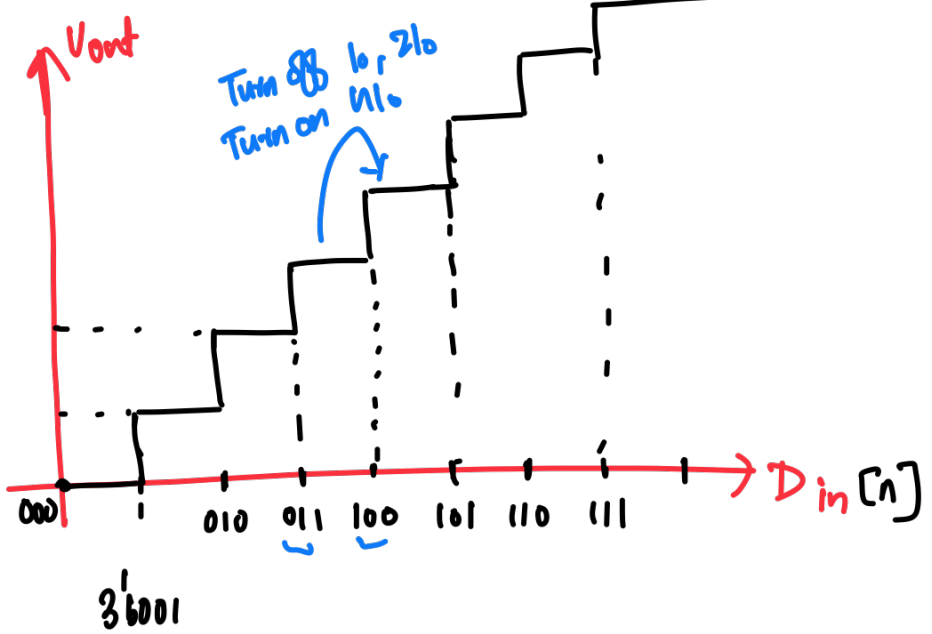
1→2 off on

ΔI_1

3→4 off off on

ΔI_2

7→8 off off off on



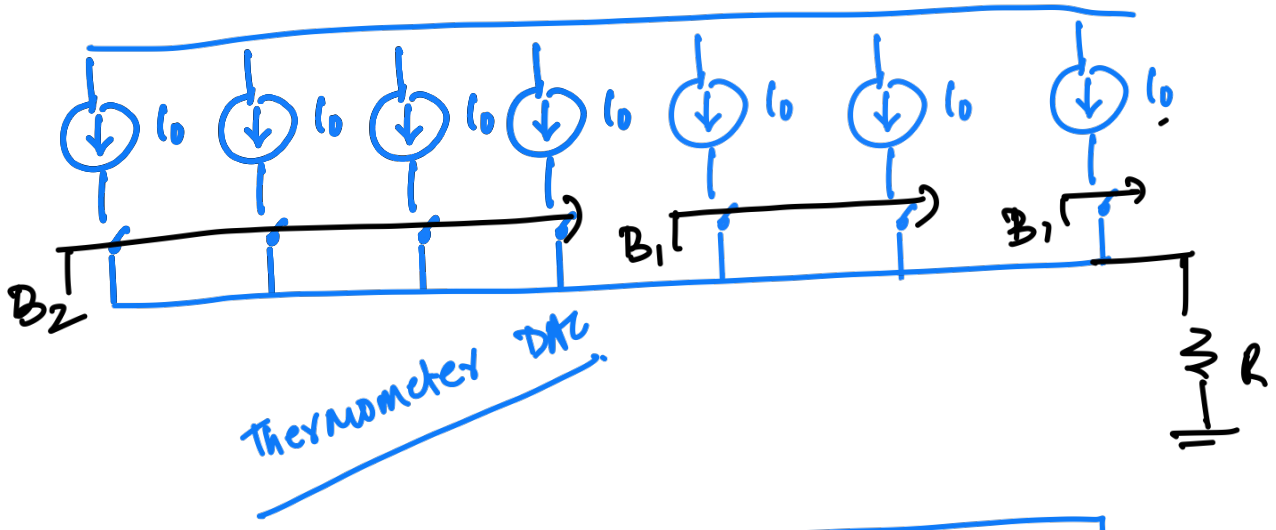
$3'b001$

3 → 4

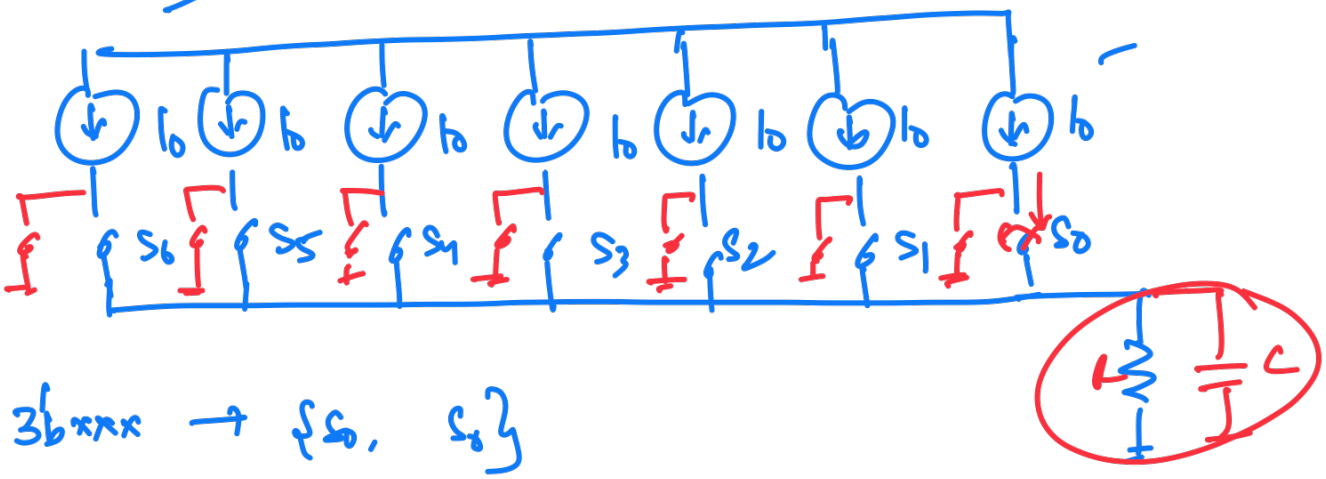
$I_{out} = (I_0+\Delta_0) + (2I_0+\Delta_1)$ → $4I_0+\Delta_2$

$\Delta_0 = \frac{\Delta_1}{2} = \frac{\Delta_2}{4}$

$\Delta_2 > \Delta_0+\Delta_1$



Thermometer DAC



$3'b\times\times\times \rightarrow \{S_0, S_6\}$

- 10-bit DAC (Segmented)

6'bit Binary DAC 4'b thermometer DAC.

